UNIQUENESS OF THE LIFT ALONG DISCRETELY VALUED DIAGRAMS IMPLIES SEPARATEDNESS OF THE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The properties of the diagonal of a morphism of algebraic spaces. Morphisms of Spaces, label spaces-morphisms-lemma-properties-diagonal (invoked at proof line 3655). It gives that Δ is representable, separated, a monomorphism and locally of finite type, which is the starting point; representability makes the fibre product V below a scheme.
- Closed immersions are detected etale locally on the target. Morphisms of Spaces, label spaces-morphisms-lemma-closed-immersion-local (invoked at proof line 3660). It reduces the closedness of Δ to the closedness of the map $V \to U$ obtained from an affine etale $U \to X \times_Y X$.
- The resulting auxiliary scheme is Noetherian, of finite type and quasi-compact. Morphisms of Spaces, labels spaces-morphisms-lemma-composition-finite-type, spaces-morphisms-lemma-base-change-finite-type and spaces-morphisms-lemma-locally-finite-type-locally-noetherian (invoked at proof lines 3662-3665). They make U Noetherian and, together with quasi-separatedness of f, show that $V \to U$ is a quasi-compact morphism of finite type, so that the valuative criterion applies.
- The valuative criterion of properness for a morphism of Noetherian schemes. Limits, label limits-lemma-Noetherian-dvr-valuative-proper (invoked at proof line 3681). Once the uniqueness of the dotted arrow is verified for all discrete valuation ring diagrams over U, it gives that $V \to U$ is proper.
- A proper monomorphism is a closed immersion. Etale Morphisms, label etale-lemma-characterize-closed-immersion (invoked at proof line 3684). Since Δ is a monomorphism and has been shown proper, it is a closed immersion, i.e. f is separated.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04XV: sources/stacks-chow/tags.txt, source lines 6408–6408. Navigation pointer for \ref{etale-lemma-characterize-closed-immersion} used inside the counted span (line 3685); the argument that consumes it is retained in the span.

Source reference. Original lemma 0208: sources/stacks-chow/tags.txt, source lines 2625–2625. Navigation pointer for \ref{limits-lemma-Noetherian-dvr-valuative-proper} used inside the counted span (line 3681); the argument that consumes it is retained in the span.

Source reference. Original lemma 03XH: sources/stacks-chow/tags.txt, source lines 5148–5148. Navigation pointer for \ref{spaces-morphisms-lemma-base-change-finite-type} used inside the counted span (line 3664); the argument that consumes it is retained in the span.

Source reference. Original lemma 03M4: sources/stacks-chow/tags.txt, source lines 4736–4736. Navigation pointer for \ref{spaces-morphisms-lemma-closed-immersion-local} used inside the counted span (line 3660); the argument that consumes it is retained in the span.

Source reference. Original lemma 03XG: sources/stacks-chow/tags.txt, source lines 5147–5147. Navigation pointer for \ref{spaces-morphisms-lemma-composition-finite-type} used inside the counted span (line 3663); the argument that consumes it is retained in the span.

Source reference. Original lemma 04ZK: sources/stacks-chow/tags.txt, source lines 6468–6468. Navigation pointer for \ref{spaces-morphisms-lemma-locally-finite-type-locally-noetherian} used inside the counted span (line 3665); the argument that consumes it is retained in the span.

Source reference. Original lemma 03HK: sources/stacks-chow/tags.txt, source lines 4570–4570. Navigation pointer for \ref{spaces-morphisms-lemma-properties-diagonal} used inside the counted span (line 3655); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume*

(1) *Y is locally Noetherian,*
(2) *f is locally of finite type and quasi-separated,*
(3) *for every commutative diagram*

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

where A is a discrete valuation ring and K its fraction field, there is at most one dotted arrow making the diagram commute.

Then f is separated.

Proof. We have to show that the diagonal $\Delta : X \to X \times_Y X$ is a closed immersion. We already know Δ is representable, separated, a monomorphism, and locally of finite type, see Morphisms of Spaces, Lemma 03HK. Choose an affine scheme U and an étale morphism $U \to X \times_Y X$. Set $V = X \times_{\Delta, X \times_Y X} U$. It suffices to show that $V \to U$ is a closed immersion (Morphisms of Spaces, Lemma 03M4). Since $X \times_Y X$ is locally of finite type over Y we see that U is Noetherian (use Morphisms of Spaces, Lemmas 03XG, 03XH, and 04ZK). Note that V is a scheme

as Δ is representable. Also, V is quasi-compact because f is quasi-separated. Hence $V \to U$ is of finite type. Consider a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & V \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & U \end{array}$$

of morphisms of schemes where A is a discrete valuation ring with fraction field K. We can interpret the composition $\mathrm{Spec}(A) \to U \to X \times_Y X$ as a pair of morphisms $a, b : \mathrm{Spec}(A) \to X$ agreeing as morphisms into Y and equal when restricted to $\mathrm{Spec}(K)$. Hence our assumption (3) guarantees $a = b$ and we find the dotted arrow in the diagram. By Limits, Lemma 0208 we conclude that $V \to U$ is proper. In other words, Δ is proper. Since Δ is a monomorphism, we find that Δ is a closed immersion (Étale Morphisms, Lemma 04XV) as desired. □